SS-B-755a
March 9, 1967

SUPERSEDING
Fed. Spec. SS-B-755
June 28, 1963 and
Fed. Spec. SS-B-750b
November 23, 1960

FEDERAL SPECIFICATION

BUILDING BOARD, ASBESTOS-CEMENT: FLAT AND CORRUGATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flat and corrugated asbestos-cement building boards.

1.2 Classification.

1.2.1 *Types.* The asbestos-cement building boards shall be of the following types, as specified (see 6.2):

A—Standard corrugated.
B—Lightweight corrugated.
F—Flexible flat.
U—Utility flat.

1.2.2 *Sizes.* Dimensions of the asbestos-cement building boards shall be in accordance with table I, as specified (see 6.2).

TABLE I. *Sizes*

Type	Dimensions (inches)		
	Thickness	Width	Length
A	3/8	42	48, 96, 120, 144
B	3/16	42	48, 96, 120, 144
F	1/8, 3/16, 1/4, 3/8, 1/2, 5/8	48	48, 96, 120, 144
U	3/16, 1/4, 3/8, 1/2, 5/8	48	48, 96, 120, 144

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-C-650—Crates, Wood, Open and Covered.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Boards shall be either corrugated or flat sheets formed under pressure and cured to meet the physical requirements specified herein.

3.2 Material. Boards and supplementary shapes, except filler strips (see 3.8.2), shall be composed of a combination of asbestos fiber and portland cement, portland blast-furnace cement, or portland-pozzalana cement, with or without the addition of curing agents, water-repellent substances, mineral fillers, coatings or pigments. Organic fiber content of the asbestos-cement combination shall be not more than 1 percent by weight.

3.3 Finish. Board surfaces intended for exposure shall be smooth.

3.4 Color. Unless otherwise specified (see 6.2), color shall be cement gray.

3.5 Dimensional tolerances.

3.5.1 *Thickness.* Board thickness shall conform to table II.

TABLE II. *Board thicknesses*

Nominal thickness	Types A and B, minimum		Types F and U	
	At crest and vale	Midway between crest and vale	Minimum	Maximum
Inch	*Inch*	*Inch*	*Inch*	*Inch*
1/8 (0.125)	—	—	0.100	0.150
3/16 (.187)	0.171	0.140	.162	.212
1/4 (.250)	—	—	.225	.275
3/8 (.375)	.343	.281	.345	.405
1/2 (.500)	—	—	.470	.530
5/8 (.625)	—	—	.595	.655

3.5.1.1 *Overall thickness between corrugations.* Overall thickness between two planes contacting the crests of corrugations on opposite sides of corrugated board shall conform to table III.

TABLE III. *Overall thickness*

Type	Overall thickness	
	Minimum	Maximum
	Inches	*Inches*
A	1-23/64	1-9/16
B	1-3/16	1-25/64

3.5.2 *Width.* Types A and B boards shall not vary from the specified width by more than plus or minus 1/4 inch. Types F and U boards shall not vary from the specified width by more than plus or minus 1/16 inch.

3.5.3 *Length.* Types A and B boards shall not vary from the specified length by more than plus or minus 1/4 inch. Types F and U boards shall not vary from the specified length by more than plus or minus 1/8 inch.

3.6 Water absorption. The average water absorption of the board, by weight, shall be not more than the following values:

Type	Percent by weight
A	25
B	25
F	25
U	30

3.7 Flexural strength. When loaded in the center as simple beams over the span distances specified, the minimum average breaking load of test specimens of board shall be not less than as shown in table IV.

TABLE IV. *Minimum breaking load*

Nominal thickness	Minimum average breaking load			
	Over 30-inch span*		Over 10-inch span	
	Type A	Type B	Type F	Type U
(Inch)	*(Pounds)*	*(Pounds)*	*(Pounds)*	*(Pounds)*
1/8	—	—	20	—
3/16	—	200	50	35
1/4	—	—	90	65
3/8	500	—	190	145
1/2	—	—	360	260
5/8	—	—	560	400

*These values shall be calculated to a width of 1 foot.

3.8 Corrugated board, types A and B. Corrugations of both types of board shall have a pitch of 4.2 inches, center to center (10 corrugations per board width).

3.8.1 *Supplementary shapes.* When specified (see 6.2), supplementary shapes of

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the following forms shall be furnished with corrugated boards in the quantities indicated.

3.8.1.1 *Half round ridge roll.* Half round ridge roll shall be from 7 to 8 inches in diameter and shall weigh between 2-1/2 to 3-3/4 pounds per lineal foot. Lengths shall be as specified (see 6.2), and battens shall be furnished to fit each piece.

3.8.1.2 *Two-piece ridge roll.* Two-piece ridge roll shall be adjustable to the roof pitch and shall be furnished in the lengths specified (see 6.2).

3.8.1.3 *Corner angles or corner rolls.* Corner angles or corner rolls shall be either inside or outside forms as specified (see 6.2), with or without overlapping bell ends. Wings shall be between 6 and 8 inches in length and the corner radius shall be from 2 to 3 inches at 90°. Weight of the angles or rolls shall be from 2-3/4 to 3-1/2 pounds per lineal foot and lengths shall be as specified (see 6.2).

3.8.1.4 *Louver blades.* Louver blades shall be 2-1/2 corrugations wide and of the lengths specified (see 6.2).

3.8.2 *Filler strips.* Filler strips for closing surfaces to corrugations shall be 37-3/4 inches long and shall be either ungrooved or grooved, as specified (see 6.2).

3.8.2.1 *Ungrooved filler strips.* Ungrooved filler strips shall be a minimum of 1 inch wide and tapered from 5/8 to 1/4 inch thick between the flat surface plane and the corrugation vale.

3.8.2.2 *Grooved filler strips.* Grooved filler strips shall be 1-1/2 to 2 inches wide and tapered from 1-1/8 to 3/4 inch thick between the flat surface plane and the corrugation vale. The flat surface plane shall be grooved to accept the louver edge of ridge roll.

3.9 Flat board, types F and U. The board shall be completely flat and the deviation from squareness of the boards shall not exceed 1/64 inch per foot of width or length.

3.9.1 *Deflection.* The average minimum deflection of a 6- by 12-inch specimen of the board under maximum breaking load shall be not less than that shown in table V.

TABLE V. *Average minimum deflection*

Nominal thickness	Type F		Type U	
	Parallel w/length	Parallel w/width	Parallel w/length	Parallel w/width
(Inch)	*(Inch)*	*(Inch)*	*(Inch)*	*(Inch)*
1/8	0.45	0.35	—	—
3/16	.30	.20	0.20	0.15
1/4	.20	.15	.15	.10
3/8	.12	.10	.10	.08
1/2	.08	.06	.07	.05
5/8	.05	.03	.04	.02

3.10 Workmanship. The boards and supplementary shapes shall be free of spalls, cracks, broken edges, broken corners, gouges, or any defects that will impair appearance, erection, use, or serviceability. Boards shall be whole and clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 *Sampling.*

4.3.1.1 *Lot.* A lot shall consist of all of the building board, supplementary shapes, or filler strips of the same type and size, offered for delivery at one time.

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4.3.1.2 *For examination.* Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.3 *For tests.* Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-1.

4.3.2 *Examination.* Samples selected in accordance with 4.3.1.2 shall be examined as specified in 4.4.1. Acceptable quality level (AQL) shall be 2.5 percent defective.

4.3.3 *Tests.* Samples selected in accordance with 4.3.1.3 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 **Inspection procedure.**

4.4.1 *Examination.* The board and supplementary shapes shall be examined for the following defects:

101. Material not as specified.
102. Surface intended for exposure not smooth.
103. Color not as specified.
104. Width not as specified.
105. Length not as specified.
106. Corrugations not as specified (types A and B).
107. Supplementary shapes not as specified (types A and B).
108. Battens missing from half round ridge rolls.
109. Filler strips not as specified (types A and B).
110. Flat boards not square within plus or minus 1/64 inch per foot of width or length (types F and U).
111. Workmanship not as specified.

4.4.2 *Test methods.*

4.4.2.1 *Preparation of specimens.* Cut two specimens from each board in the sample and from the interior area of the board so that no specimen edge is cut less than 3 inches from the original board edge, except for flexural strength specimens of corrugated board which must be at least 4.2 inches from the original board edge. Specimen size shall be 6 ± 1/16 inch in width and 12 ± 1/16 inch in length, except for flexural strength specimens of corrugated board which shall be 12-1/2 ± 1/16 inch in width and 36 ± 1/8 inch in length. One specimen shall be cut with the long edge parallel to the length of the board and the other shall be cut with the long edge parallel to the width of the board, except that flexural strength specimens of corrugated board shall have the long edge of the specimens parallel to the corrugations. Dry each specimen to a constant weight in a ventilated oven at a temperature of 212° to 220°F. cool to room temperature in a dessicator, record the dry weight of each specimen, and store in the dessicator.

4.4.2.2 *Thickness.* Measure the thickness of flat board specimens using a micrometer caliper having flat anvils 1/4 inch in diameter and reading to 0.001 inch. Make four measurements on each specimen at the midpoint of each edge, and at least 1/2 inch in from the edge. Measure the thickness of corrugated specimens using a micrometer having a ball anvil and ball spindle ends 1/8 to 1/4 inch in diameter, or an anvil and spindle, each with a 30° taper and 1/64-inch flat at its end. Measure the thickness of the corrugated specimens at the crest, the vale, and at the flank, midway between the crest and vale. The thickness of the crest and vale shall be the average of four measurements, two at the crest and two at the vale, one of each at each end. The thickness at the flank shall be the average of four measurements, two at each end of the specimen. Measure overall thickness between corrugations by clamping straightedges in contact with not less than four crests of the corrugations on both top and bottom of the specimen and measuring the distance between the contacting edges. Board thicknesses not conforming to table II or III, as applicable, shall constitute failure of this test.

4.4.2.3 *Water absorption.* Weigh a broken segment of a prepared specimen (pieces may be taken from the specimens broken in flexural strength tests), and submerge for 24 ± 1/2 hour in clean water at 60° to 80°F. Remove the segment from the wa-

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ter, wipe with a damp cloth, and reweigh. Calculate the water absorption value for each segment as follows:

$$\text{Water absorption, percent by weight} = \frac{Ws - Wd}{Wd} \times 100$$

Where:

Ws = saturated weight, and
Wd = dry weight.

Average the values for all segments tested. Water absorption greater than as specified in 3.6 shall constitute failure of this test.

4.4.2.4 *Flexural strength.* Place the prepared specimen on supports that cannot exert longitudinal constraint (such as rocker-type bearing edges or rollers with a 1/8-inch minimum and a 1/2-inch maximum radius), and apply the load at the midspan through a similar edge bearing against the finished surface of the specimen. The test span shall be 30 ± 1/16 inch for types A and B boards and 10 ± 1/16 inch for types F and U boards. The load line and supports shall be parallel. Mount a dial micrometer reading to 0.01 inch to bear on the specimen at the load line when testing types F and U boards in order to record the deflection. Increase the load at a uniform rate which will induce failure within 1 to 2 minutes. Calculate the flexural strength as the average load in pounds at which all specimens from one sample failed. The error in the load reading shall not exceed 1 percent at maximum (breaking) load. Types A and B flexural strengths should be calculated to a width of 1 foot. Inability of the sample to attain an average minimum flexural strength as specified in table IV shall constitute failure of this test.

4.4.2.5 *Deflection.* Record the depth of deflection of each specimen at failure during the flexural strength tests and calculate the average deflection of all the specimens from one sample. An average deflection at maximum load which is less than the value specified in table V shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 *Quality conformance inspection of pack.*

4.5.1.1 *Unit of product.* For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 *Sampling.* Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-2.

4.5.1.3 *Examination.* Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Materials and containers not as specified for levels A and B. Each incorrect material or container shall be considered one defect.
113.	Crate containing building board exceeds 3,000 pounds for level A.
114.	Strapping not zinc coated for level A.
115.	Supplementary shapes and filler strips not blocked and braced within the box to prevent movement for levels A and B.
116.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Sheets of building board of like description shall be uniformly packed in a crate conforming to PPP-C-650, style A, type II. The weight limitation of the crate shall be waived; however, the gross weight shall not exceed 3,000 pounds. Strapping shall be zinc coated.

5.1.1.1 *Supplementary shapes and filler strips.* Supplementary shapes and filler strips shall be packed in boxes conforming to PPP-B-601, overseas type, style optional. The contents shall be blocked and braced within the box to prevent movement. Strapping shall be zinc coated.

Note: The supplementary shapes and filler strips shall be shipped separately from, but at the same time, as the crated sheets.

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5.1.2 *Level B.* Packing shall be as specified for level A, except strapping for crates and boxes need not be zinc coated and boxes for supplementary shapes and filler strips shall be domestic type.

5.1.3 *Level C.* The sheets of building board, supplementary shapes, and filler strips shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in containers with carrier rules applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking specified in the contract or order, marking for shipment shall be in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking specified in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Asbestos-cement building boards are intended for use in low-cost, temporary, and permanent construction.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(*a*) Title, number, and date of this specification.
(*b*) Type and size required (see 1.2.1 and 1.2.2).
(*c*) Color required if other than cement gray (see 3.4).
(*d*) Supplementary shapes required (see 3.8.1), and specify:
  (1) Whether half-round or 2-piece ridge roll is required and length (see 3.8.1.1 and 3.8.1.2).
  (2) Whether inside or outside corner angle or roll is required and length (see 3.8.1.3).
  (3) Length of louver blade required (see 3.8.1.4).
  (4) Whether filler strips shall be grooved or ungrooved (see 3.8.2).
(*e*) Selection of applicable level of packing required (see 5.1).

6.3 Efflorescence. Efflorescence that sometimes appears on asbestos-cement board, is not a defect and should not result in a permanent change in color of the board.

CUSTODIANS:
Army—ME
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